Contents

Chapter 1 Classical Algebra

Exercise 1.2

Assume for contradiction that there exists $a, b \in \mathbb{Z}$ with $b \neq 0$ such that $\left(\frac{a}{b}\right)^2 = 2$. If a is negative and b is positive, then replacing a with $-a$ gives us $\left(\frac{-a}{b}\right)^2 = \left(\frac{a}{b}\right)^2 = 2$. Therefore, $-a, b$ is also a valid pair of integers satisfying the conditions. If b is negative and a is positive, then replacing b with $-b$ gives us $\left(\frac{a}{-b}\right)^2 = \left(\frac{a}{b}\right)^2 = 2$. Therefore, $a, -b$ is also a valid pair of integers satisfying the conditions. If both a and b are negative, then replacing a with $-a$ and b with $-b$ gives us $\left(\frac{-a}{-b}\right)^2 = \left(\frac{a}{b}\right)^2 = 2$. Therefore, $-a, -b$ is also a valid pair of integers satisfying the conditions. Hence, we can assume without loss of generality that $a, b > 0$.

Since we assume that b only takes values in $\mathbb{N}$, by the well-ordering principle, there exists a smallest b satisfying the conditions. Note that

$$\begin{aligned}
\left(\frac{2b-a}{a-b}\right)^2 &= \frac{4b^2 - 4ab + a^2}{a^2 - 2ab + b^2} \\
&= \frac{4 - 4\left(\frac{a}{b}\right) + \left(\frac{a}{b}\right)^2}{\left(\frac{a}{b}\right)^2 - 2\left(\frac{a}{b}\right) + 1} \\
&= \frac{4 - 4\left(\frac{a}{b}\right) + 2}{2 - 2\left(\frac{a}{b}\right) + 1} \\
&= \frac{6 - 4\left(\frac{a}{b}\right)}{3 - 2\left(\frac{a}{b}\right)} \\
&= \frac{6 - 4\left(\frac{a}{b}\right)}{3 - 2\left(\frac{a}{b}\right)} \times \frac{3 + 2\left(\frac{a}{b}\right)}{3 + 2\left(\frac{a}{b}\right)} \\
&= \frac{18 + 12\left(\frac{a}{b}\right) - 12\left(\frac{a}{b}\right) - 8\left(\frac{a}{b}\right)^2}{9 - 4\left(\frac{a}{b}\right)^2} \\
&= \frac{18 - 8 \times 2}{9 - 4 \times 2} \\
&= 2.
\end{aligned}$$

Recall that $\left(\frac{a}{b}\right)^2 = 2$, so we have $a^2 = 2b^2$ and thus $2b^2 - a^2 = 0$. Note that

$$\begin{aligned}
(2b-a)(2b+a) &= 4b^2 - a^2 \\
&= 2b^2 + (2b^2 - a^2) \\
&= 2b^2 \\
&> 0
\end{aligned}$$

since $b > 0$ by assumption. Since $a, b > 0$, we have $2b + a > 0$. Hence, $2b - a > 0$. Also, from $a^2 = 2b^2$ we have $a^2 - b^2 = b^2$. Considering $b > 0$, we have

$$\begin{aligned}
b^2 &> 0 \\
a^2 - b^2 &> 0 \\
(a-b)(a+b) &> 0 \\
a - b &> 0
\end{aligned}$$

since $a+b>0$. From $2b-a>0$ and $a-b>0$ we know that $2b-a, a-b$ is also a pair of positive integers satisfying the conditions. However, from $2b-a>0$ we have $a<2b$, and hence $a-b<2b-b=b$, contradicting the minimality of b. $\square$

Exercise 1.5

Let $S=\{p+q\alpha+r\alpha^2 \mid p,q,r\in\mathbb{Q}\}$. We first prove that S is a subring of $\mathbb{C}$.

Note that $1=1+0\alpha+0\alpha^2\in S$. Let $p+q\alpha+r\alpha^2, p'+q'\alpha+r'\alpha^2\in S$, where $p,q,r,p',q',r'\in\mathbb{Q}$. Then we have $(p+q\alpha+r\alpha^2)+(p'+q'\alpha+r'\alpha^2)=(p+p')+(q+q')\alpha+(r+r')\alpha^2\in S$, $-(p+q\alpha+r\alpha^2)=(-p)+(-q)\alpha+(-r)\alpha^2\in S$ and $(p+q\alpha+r\alpha^2)(p'+q'\alpha+r'\alpha^2)=(pp'+2rq')+(pq'+p'q+2rr')\alpha+(pr'+qq'+qr'+rp')\alpha^2\in S$. By the subring test, S is a ring.

Next, we construct an inverse explicitly to show that S is a subfield. Let $x=p+q\alpha+r\alpha^2\in S\backslash\{0\}$, where $p,q,r\in\mathbb{Q}$. Then we have $x^2=(p^2+4qr)+(2pq+2r^2)\alpha+(2pr+q^2)\alpha^2$ and $x^3=(p^3+12pqr+2q^3+4r^3)+(3p^2q+6pr^2+6q^2r)\alpha+(3p^2r+3pq^2+6qr^2)\alpha^2$. Let $x^3+k_2x^2+k_1x+k_0=0$. Denote the coefficient of α^j in x^i by x_{ij}. Then we have $(x_{30}+x_{31}\alpha+x_{32}\alpha^2)+k_2(x_{20}+x_{21}\alpha+x_{22}\alpha^2)+k_1(p+q\alpha+r\alpha^2)+k_0=0$, that is, $(x_{30}+x_{20}k_2+pk_1+k_0)+(x_{31}+x_{21}k_2+qk_1)\alpha+(x_{32}+x_{22}k_2+rk_1)\alpha^2=0$. Since $1,\alpha,\alpha^2$ are linearly independent, comparing coefficients gives us the linear system

$$\begin{cases} k_0+pk_1+x_{20}k_2=-x_{30} \\ qk_1+x_{21}k_2=-x_{31} \\ rk_1+x_{22}k_2=-x_{32} \end{cases}.$$

We can use the following python code to solve this system.

```python
import sympy as sp

k0, k1, k2 = sp.symbols('k0 k1 k2')
p, q, r = sp.symbols('p q r')

x30 = p**3 + 12 * p * q * r + 2 * (q**3) + 4 * (r**3)
x31 = 3 * (p**2) * q + 6 * p * (r**2) + 6 * (q**2) * r
x32 = 3 * (p**2) * r + 3 * p * (q**2) + 6 * q * (r**2)
x20 = p**2 + 4 * q * r
x21 = 2 * p * q + 2 * (r**2)
x22 = 2 * p * r + q**2

M = sp.Matrix([
    [1, p, x20, - x30],
    [0, q, x21, - x31],
    [0, r, x22, - x32]
])

solution = sp.solve_linear_system(M, k0, k1, k2)

for var in [k0, k1, k2]:
    print(f"{var} =")
    sp.pprint(sp.simplify(solution[var]))
```

The solution is $k_0=-p^3+6pqr-2q^3-4r^3$, $k_1=3p^2-6qr$ and $k_2=-3p$. Since $x^3+k_2x^2+k_1x+k_0=0$, we have $x(x^2+k_2x+k_1+k_0x^{-1})=0$. Recall that $\mathbb{C}$ is a field and hence a domain, and $x \neq 0$. Therefore, $x^2 + k_2 x + k_1 + k_0 x^{-1} = 0$, that is,

$$x^{-1} = -\frac{1}{k_0}x^2 - \frac{k_2}{k_0}x - \frac{k_1}{k_0}.$$

Since x and x^2 are linear combinations of $1, \alpha, \alpha^2$, x^{-1} is also a linear combination of $1, \alpha, \alpha^2$. $\square$

Chapter 2 The Fundamental Theorem of Algebra

Exercise 2.1

We first prove the existence of such an expression. If $p(t)$ is constant, then set $p(t) = q(t)$ and this is the required form, since a constant polynomial $q(t)$ has no zeros in $\mathbb{Q}[t]$. If $p(t)$ is not constant, then $\partial p \geq 1$. By Proposition 2.7, there exist $\alpha_1, \dots, \alpha_n \in \mathbb{C}$ and $0 \neq k \in \mathbb{C}$ such that $p(t) = k(t - \alpha_1)\dots(t - \alpha_n)$. The coefficient of t^n in the expression on the right hand side is k. Since $p(t) \in \mathbb{Q}[t]$, we have $k \in \mathbb{Q}$. Assume without loss of generality that $\alpha_1, \dots, \alpha_r \in \mathbb{Q}$ and $\alpha_{r+1}, \dots, \alpha_n \notin \mathbb{Q}$ for some $0 \leq r \leq n$. Let $q(t) = k(t - \alpha_{r+1})\dots(t - \alpha_n)$. Then $p(t) = (t - \alpha_1)\dots(t - \alpha_r)q(t)$. Since $\alpha_{r+1}, \dots, \alpha_n$ are the only complex roots of $q(t)$, $q(t)$ has no roots in $\mathbb{Q}[t]$.

Suppose

$$p(t) = (t - \alpha_1)\dots(t - \alpha_r)q(t) = (t - \hat{\alpha}_1)\dots(t - \hat{\alpha}_s)\hat{q}(t)$$

such that $\alpha_1, \dots, \alpha_r, \hat{\alpha}_1, \dots, \hat{\alpha}_s \in \mathbb{Q}$ and $q(t), \hat{q}(t)$ have no zeros in $\mathbb{Q}[t]$. Assume without loss of generality that $\alpha_1 \leq \dots \leq \alpha_r$. Substitute $t = \alpha_1$, we get

$$0 = (\alpha_1 - \alpha_1)\dots(\alpha_1 - \alpha_r)q(\alpha_1) = (\alpha_1 - \hat{\alpha}_1)\dots(\alpha_1 - \hat{\alpha}_s)\hat{q}(\alpha_1).$$

Since $\hat{q}(\alpha_1) \neq 0$, we have $\alpha_1 - \hat{\alpha}_1 = 0$ or $\alpha_1 - \hat{\alpha}_2 = 0$ or $\dots$ or $\alpha_1 - \hat{\alpha}_s = 0$. Hence, $\alpha_1 = \hat{\alpha}_1$ or $\alpha_1 = \hat{\alpha}_2$ or $\dots$ or $\alpha_1 = \hat{\alpha}_s$. Similarly, by substituting $t = \alpha_2, \dots, \alpha_r$, we have $\alpha_1, \alpha_2, \dots, \alpha_r \in \{\hat{\alpha}_1, \hat{\alpha}_2, \dots, \hat{\alpha}_s\}$. Assume without loss of generality that $\alpha_1 = \hat{\alpha}_1$, $\alpha_2 = \hat{\alpha}_2$, $\dots$, $\alpha_r = \hat{\alpha}_r$. Assume for contradiction that $s > r$, so $\hat{\alpha}_{r+1}, \dots, \hat{\alpha}_s$ exist. Since $(t - \alpha_1)\dots(t - \alpha_r) = (t - \hat{\alpha}_1)\dots(t - \hat{\alpha}_r) \neq \mathbf{0}$, we have $q(t) = (t - \hat{\alpha}_{r+1})\dots(t - \hat{\alpha}_s)\hat{q}(t)$. Note that $\hat{\alpha}_{r+1}$ is a rational root of the polynomial on the right hand side, contradicting that $q(t)$ has no rational roots. Hence, $\hat{\alpha}_{r+1}, \dots, \hat{\alpha}_s$ do not exist and thus $q(t) = \hat{q}(t)$. This proves the uniqueness of such an expression.

Note that $p(\alpha_j) = (\alpha_j - \alpha_1)\dots(\alpha_j - \alpha_j)\dots(\alpha_j - \alpha_r)q(\alpha_j) = 0$, so α_j is a zero of $p(t)$ in $\mathbb{Q}$. Conversely, we want to prove that if $x \in \mathbb{Q}$ and $p(x) = 0$ then $x = \alpha_j$ for some j. Assume for contradiction that there exists $x \in \mathbb{Q}$ such that $p(x) = 0$ and $x \neq \alpha_j$ for all j. Then we have $x - \alpha_j \neq 0$ for all j. Since $q(t)$ has no zeros in $\mathbb{Q}[t]$, we have $q(x) \neq 0$. Since $\mathbb{Q}$ is a domain, we have $p(x) = (x - \alpha_1)\dots(x - \alpha_r)q(x) \neq 0$, which contradicts that x is a root of $p(t)$. Therefore, the α_j are precisely the zeros of $p(t)$ in $\mathbb{Q}$. $\square$

Exercise 2.6

Let $f(t)$ be a polynomial with $\partial f = n$. Suppose $f(t) = a_0 + a_1 t + a_2 t^2 + \dots + a_n t^n = 0$ for all $t \in \mathbb{C}$. Since $f(1) = 0$, we have

$$a_0 + a_1 + a_2 + \dots + a_n = 0.$$

Since $f(2) = 0$, we have

$$a_0 + 2a_1 + 2^2 a_2 + \ldots + 2^n a_n = 0.$$

Similarly, since $f(k) = 0$ for any $1 \leq k \leq n+1$, we have

$$a_0 + ka_1 + k^2 a_2 + \ldots + k^n a_n = 0.$$

Hence, we have the linear system

$$\begin{pmatrix} 1 & 1 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 2^2 & 2^3 & \cdots & 2^n \\ 1 & 3 & 3^2 & 3^3 & \cdots & 3^n \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & n+1 & (n+1)^2 & (n+1)^3 & \cdots & (n+1)^n \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Note that

$$\begin{vmatrix} 1 & 1 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 2^2 & 2^3 & \cdots & 2^n \\ 1 & 3 & 3^2 & 3^3 & \cdots & 3^n \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & n+1 & (n+1)^2 & (n+1)^3 & \cdots & (n+1)^n \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 3 & \cdots & n+1 \\ 1^2 & 2^2 & 3^2 & \cdots & (n+1)^2 \\ 1^3 & 2^3 & 3^3 & \cdots & (n+1)^3 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1^n & 2^n & 3^n & \cdots & (n+1)^n \end{vmatrix},$$

which is the Vandermonde determinant with $z_1 = 1$, $z_2 = 2$, …, $z_{n+1} = n+1$. By Exercise 2.5, this determinant is non-zero. So the linear system has no non-trivial solutions. Hence, $a_0 = a_1 = a_2 = \ldots = a_n = 0$. Therefore, all the coefficients of f are zero. □